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324351 (14)

BE (3rd Semester)

Examination, Nov.-Dec., 2014

Branch : Electrical Engineering

MATHEMATICS - III (NEW)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) of each question is compulsory. Attempt any two parts from reamaining three parts in each question.

Q. 1. (a) If $f(x) = \begin{bmatrix} 1+\frac{2x}{\pi}, & -\pi \le x \le 0 \\ 1-\frac{2x}{\pi}, & 0 \le x \le \pi \end{bmatrix}$

find the Fourier coefficient a_0. 2

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(b) Prove that 7

$$x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} (-1)^n \frac{\cos nx}{n^2},$$

$$-\pi < x < \pi$$

Hence show that

$$\sum \frac{1}{n^2} = \frac{\pi^2}{6}$$

(c) If $f(x) = |\cos x|$, expand $f(x)$ as a Fourier series in the interval $(-\pi, \pi)$. 7

(d) Obtain the first three coefficient in the Fourier cosine series for y where y is given in the following table : 7

x	:	0	1	2	3	4	5
y	:	4	8	15	7	6	2

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Q. 2. (a) Find the Laplace transform of

$\sin 2t \cdot \sin 3t$. **2**

(b) Find the Laplace transform of the following functions :

(i) $\dfrac{1-\cos 3t}{t}$ **4**

(ii) $t\,e^{2t}\sin 3t$ **3**

(c) Apply convolution theorem to evaluate **7**

$$L^{-1}\left\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right\}$$

(d) Solve the following equation by transform method : **7**

$$\frac{d^2y}{dt^2}+4\frac{dy}{dt}+3y=e^{-t}$$

$y(0) = y'(0) = 1.$

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Q. 3. (a) Form the partial differential equation

$z = f(x + at) + g(x - at)$ **2**

(b) Solve

$$\frac{\partial^2 z}{\partial x \partial y} = \sin x . \sin y$$

for which $\dfrac{\partial z}{\partial y} = -2\sin y$ when $x = 0$ and $z = 0$ when y is a odd multiple of $\dfrac{\pi}{2}$. **7**

(c) Solve **7**

$$\frac{\partial^2 z}{\partial x^2}+2\frac{\partial^2 z}{\partial x \partial y}+\frac{\partial^2 z}{\partial y^2}=x^2+xy+y^2$$

(d) Using the method of separation of variable, solve **7**

$\dfrac{\partial u}{\partial x} = 2\dfrac{\partial u}{\partial t} + u$, where

$u(x, 0) = 6e^{-3x}$

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Q. 4. (a) Write the polar form of Cauchy-Riemann equations. 2

(b) If f(z) is a regular function of z. Prove that

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4|f'(z)|^2$$ 7

(c) Evaluate, using Cauchy's integral formulae

$$\int_C \frac{\log z}{(z-1)^3}dz$$

where C is $|z-1| = \frac{1}{2}$. 7

(d) Apply the calculus of residues, to prove that

$$\int_0^{\pi} \frac{d\theta}{17-8\cos\theta} = \frac{\pi}{15}$$ 7

Q. 5. (a) Write convolution theorem for z transform. 2

(b) Given that $z(u_n) = \frac{2z^2+3z+4}{(z-3)^2}$, $|z| > 3$ show that $u_1 = 2$, $u_2 = 21$, $u_3 = 139$ 7

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(c) Using inversion integral, find inverse z transform of

$$\frac{z^2+z}{(z-1)(z^2+1)}$$ 7

(d) Solve

$$y_{n+2} + 6y_{n+1} + 9y_n = 2^n$$

with $y_0 = y_1 = 0$. 7

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